

Database Systems		
Lab #:	08	
Lab Title:	Implementation of DDL commands of SQL with suitable examples	
Submitted by:		
Names		Registration #
Sumaira Safeer		FA22-BCE-019

Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Lab No:08

Title: Implementation of DDL commands of SQL with suitable examples.

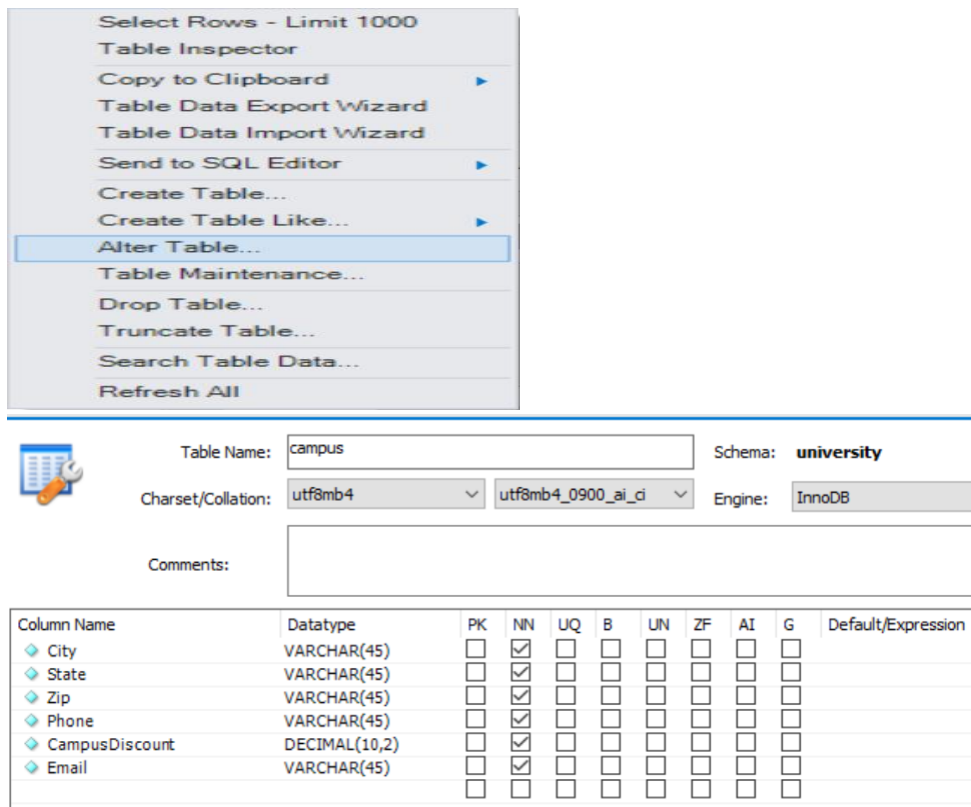
- Create table.
- Alter table.
- Drop Table.
- Rename table.

Objective:

- To understand the different issues involved in the design and implementation of a database system.
- To understand and use data definition language to write query for a database.

Task 1:

- Add the new field Email_id to the Campus table.



Select Rows - Limit 1000
Table Inspector
Copy to Clipboard
Table Data Export Wizard
Table Data Import Wizard
Send to SQL Editor
Create Table...
Create Table Like...
Alter Table...
Table Maintenance...
Drop Table...
Truncate Table...
Search Table Data...
Refresh All

Table Name: campus Schema: university
Charset/Collation: utf8mb4 utf8mb4_0900_ai_ci Engine: InnoDB
Comments:

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
City	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
State	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Zip	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Phone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
CampusDiscount	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	
Email	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	

Result Grid:

Result Grid									
	CampusID	CampusName	Street	City	State	Zip	Phone	CampusDiscount	Email
▶	2	Hannah Zoe	Main Street 4	Attock	KPK	44000	0333-6853234	45.99	HannahZoe@gmail.com
	3	Rohan Malik	Street 4	Lahore	Punjab	20544	0300-3468643	5534.87	RohanMalik@gmail.com
	4	Saamia Zee	Main Street 2	Mianwali	Punjab	22276	0343-7531435	987.43	SaamiaZee@gmail.com
	5	Subhana Hashim	Street 6	Sargodah	Punjab	22334	0344-7532685	76.64	SubhanaHashim@gmail.com
	1	Sumaira Safeer	Street 3	Bhakkar	Punjab	44022	0305-3578543	44.66	SumairaSafeer@gmail.com
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Add the new field Year_of_Joining in the Member table.

	Table Name: campus	Schema: university								
	Charset/Collation: utf8mb4 utf8mb4_0900_ai_ci	Engine: InnoDB								
Comments:										
Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
State	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Zip	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Phone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
CampusDiscount	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	
Email	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Year_of_joining	INT	☐	☒	☐	☐	☐	☐	☐	☐	

Result Grid:

Result Grid										
	CampusID	CampusName	Street	City	State	Zip	Phone	CampusDiscount	Email	Year_of_joining
	2	Hannah Zoe	Main Street 4	Attock	KPK	44000	0333-6853234	45.99	HannahZoe@gmail.com	2024
	3	Rohan Malik	Street 4	Lahore	Punjab	20544	0300-3468643	5534.87	RohanMalik@gmail.com	2022
	4	Saamia Zee	Main Street 2	Mianwali	Punjab	22276	0343-7531435	987.43	SaamiaZee@gmail.com	2023
	5	Subhana Hashim	Street 6	Sargodah	Punjab	22334	0344-7532685	76.64	SubhanaHashim@gmail.com	2020
	1	Sumaira Safeer	Street 3	Bhakkar	Punjab	44022	0305-3578543	44.66	SumairaSafeer@gmail.com	2022
	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Add the new field Brand in the OrderLine table.

	Table Name: order_line	Schema: food baker							
	Charset/Collation: utf8mb4 utf8mb4_0900_ai_	Engine: InnoDB							
Comments:									
Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G
Order_ID	INT	☒	☒	☐	☐	☐	☐	☐	☐
FoodItems_ID	INT	☒	☒	☐	☐	☐	☐	☐	☐
Quantity	INT	☐	☒	☐	☐	☐	☐	☐	☐
Brand	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐

Result Grid:

Result Grid

Filter Rows:

Edit:

Export/Import:

	Order_ID	FoodItems_ID	Quantity	Brand
▶	11	1	5	Gucci
	22	2	2	Zulbari
	33	4	6	J.
	44	3	3	Amfti
	55	5	4	GulAhmed
*	NULL	NULL	NULL	NULL

(b) ALTER TABLE...MODIFY...:

This is used to change the width as well as data type of field of existing relations.

SYNTAX: ALTER TABLE relation_name MODIFY field_1 newdata_type (Size);

1. ALTER TABLE Campus MODIFY CampusName VARCHAR(300);

Table Name: campus		Schema: university								
Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
⚡ CampusID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
🔹 CampusName	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Street	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 City	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 State	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Zip	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Phone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 CampusDiscount	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Email	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Year_of_joining	INT	☐	☒	☐	☐	☐	☐	☐	☐	

Table Name: campus		Schema: university								
Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
⚡ CampusID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
🔹 CampusName	VARCHAR(300)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Street	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 City	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 State	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Zip	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Phone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 CampusDiscount	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Email	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
🔹 Year_of_joining	INT	☐	☒	☐	☐	☐	☐	☐	☐	
campuscol	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	

2. ALTER TABLE Position MODIFY YearlyMembership FLOAT;

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
PositionID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
Position	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
YearlyMembership	DECIMAL	☐	☒	☐	☐	☐	☐	☐	☐	

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
PositionID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
Position	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
YearlyMembership	FLOAT	☐	☒	☐	☐	☐	☐	☐	☐	

Task 2:

- Change the datatype and size of Position from char(20) to varchar(15) in Position table.

Limit to 1000 rows

```

1 • SELECT * FROM university.position;
2
3 • ALTER TABLE position
4   MODIFY position VARCHAR(15);
5

```

2 22:10:22 SELECT * FROM university.position LIMIT 0, 1000
5 row(s) returned
0.000 sec / 0.000 sec

3 22:10:22 ALTER TABLE Position MODIFY Position VARCHAR(15)
5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
0.110 sec

order_line campus position position - Table x

Table Name: position
Schema: university

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
PositionID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
Position	VARCHAR(15)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
YearlyMembership	FLOAT	☐	☒	☐	☐	☐	☐	☐	☐	

- Change the datatype and size of Zip from Varchar(10) to int (15) in Campus table.

Limit to 1000 rows

```

1 • SELECT * FROM university.campus;
2 • ALTER TABLE campus
3   MODIFY Zip INT(15);
4

```

6 22:15:54 ALTER TABLE Campus MODIFY Zip INT
5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
0.047 sec

9 22:20:44 SELECT * FROM university.position LIMIT 0, 1000
5 row(s) returned
0.000 sec / 0.000 sec

Database Systems (CSC371) By: Sumaira Safeer

5

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
CampusID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
CampusName	VARCHAR(300)	☐	☒	☐	☐	☐	☐	☐	☐	
Street	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
City	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
State	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Zip	INT(15)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
Phone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
CampusDiscount	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	
Email	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	
Year_of_joining	INT	☐	☒	☐	☐	☐	☐	☐	☐	

- Change the datatype and size of First_name form char(10) to varchar(20) Members table.

lab8_task3. position position - Table campus campus - Table member x

Limit to 1000 rows

```

1 • SELECT * FROM university.member;
2 • ALTER TABLE member
3   MODIFY FirstName VARCHAR(20);
4
5

```

10 22:21:58 SELECT * FROM university.member LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec

Table Name: member Schema: university

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
MemberID	INT	☒	☒	☐	☐	☐	☐	☐	☐	
FirstName	VARCHAR(20)	☐	☒	☐	☐	☐	☐	☐	☐	
LastName	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Address	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
CampusPhone	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
CampusID	INT	☐	☒	☐	☐	☐	☐	☐	☐	
PositionID	INT	☐	☒	☐	☐	☐	☐	☐	☐	
ContractDuration	INT	☐	☒	☐	☐	☐	☐	☐	☐	

Task 3:

- Remove the column Email from campus table.

Limit to 1000 rows

```

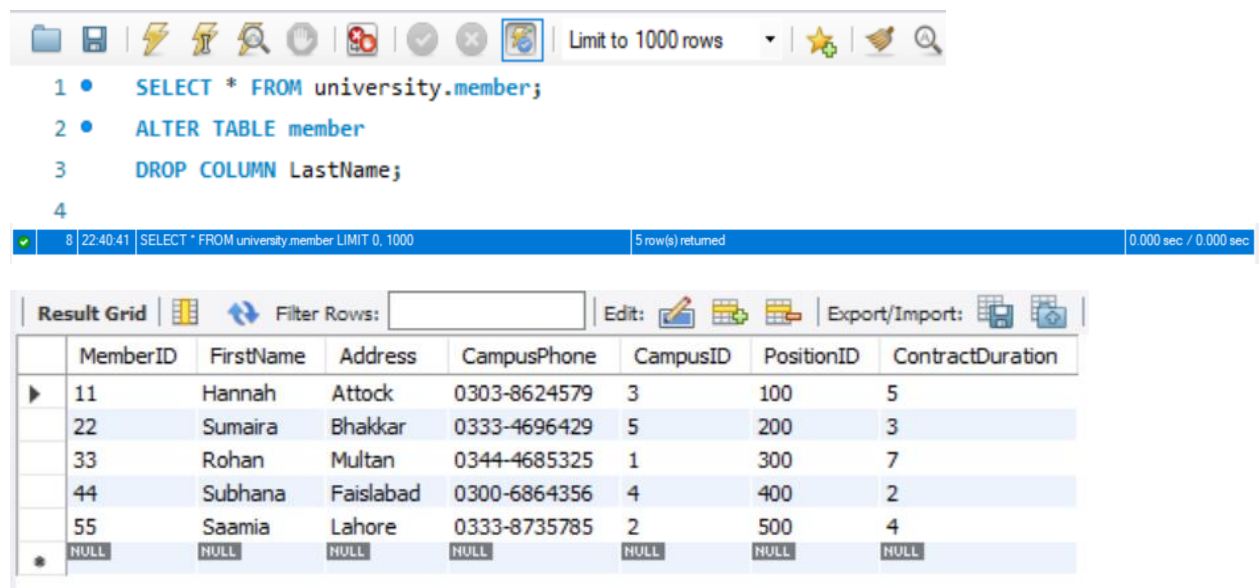
1 • SELECT * FROM university.campus;
2 • ALTER TABLE campus
3   DROP COLUMN Email;
4

```

result Grid Filter Rows: Edit Export/Import: Wrap Cell Content:

CampusID	CampusName	Street	City	State	Zip	Phone	CampusDiscount	Year_of_joining
1	Sumaira Safeer	Street 3	Bhakkar	Punjab	44022	0305-3578543	44.66	2022
2	Hannah Zoe	Main Street 4	Attock	KPK	44000	0333-6853234	45.99	2024
3	Rohan Malik	Street 4	Lahore	Punjab	20544	0300-3468643	5534.87	2022
4	Saamia Zee	Main Street 2	Mianwali	Punjab	22276	0343-7531435	987.43	2023
5	Subhana Hashim	Street 6	Sargodah	Punjab	22334	0344-7532685	76.64	2020
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Remove the field LastName from member table.



```

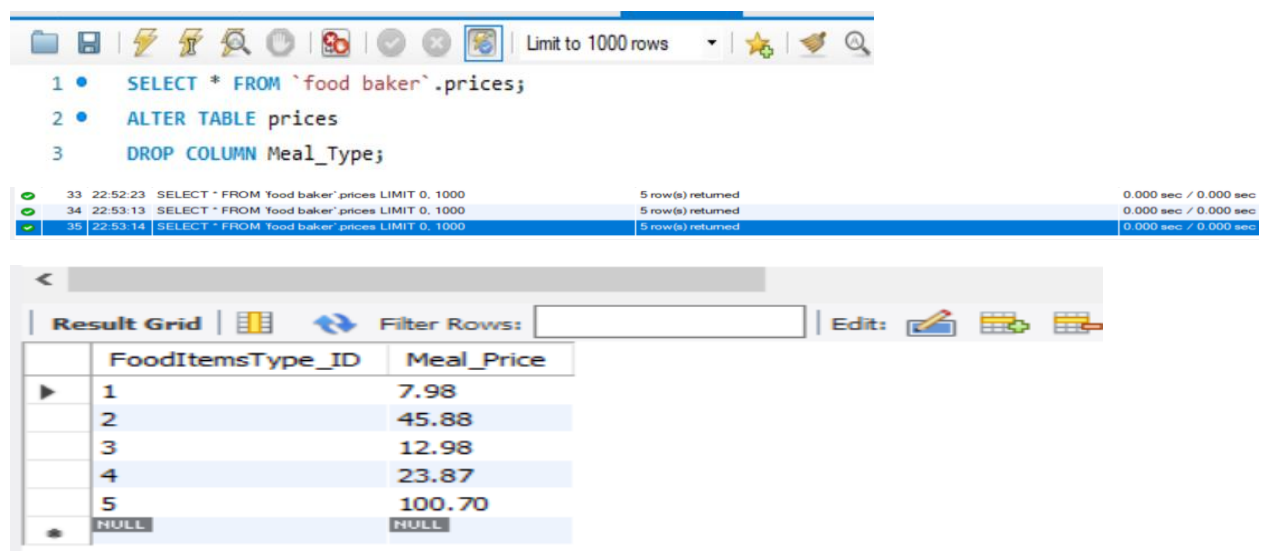
1 • SELECT * FROM university.member;
2 • ALTER TABLE member
3   DROP COLUMN LastName;
4

```

8 | 22:40:41 | SELECT * FROM university.member LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec

MemberID	FirstName	Address	CampusPhone	CampusID	PositionID	ContractDuration
11	Hannah	Attock	0303-8624579	3	100	5
22	Sumaira	Bhakkar	0333-4696429	5	200	3
33	Rohan	Multan	0344-4685325	1	300	7
44	Subhana	Faislabad	0300-6864356	4	400	2
55	Saamia	Lahore	0333-8735785	2	500	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL

- Remove the field Meal_Type form prices Table.



```

1 • SELECT * FROM 'food baker'.prices;
2 • ALTER TABLE prices
3   DROP COLUMN Meal_Type;

```

33 | 22:52:23 | SELECT * FROM 'food baker'.prices LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec
34 | 22:53:13 | SELECT * FROM 'food baker'.prices LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec
35 | 22:53:14 | SELECT * FROM 'food baker'.prices LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec

FoodItemsType_ID	Meal_Price
1	7.98
2	45.88
3	12.98
4	23.87
5	100.70
NULL	NULL

Task 4:

DROP TABLE: This is used to delete the structure of a relation. It permanently deletes the records in the table.

SYNTAX: DROP TABLE relation_name;

- Change the column name CampusName to C_name in Campus table.

Limit to 1000 rows

```

1 • SELECT * FROM university.campus;
2
3 • ALTER TABLE campus
4   RENAME COLUMN CampusName TO C_name;
5

```

CampusID	C_name	Street	City	State	Zip	Phone	CampusDiscount	Year_of_joining
1	Sumaira Safeer	Street 3	Bhakkar	Punjab	44022	0305-3578543	44.66	2022
2	Hannah Zoe	Main Street 4	Attock	KPK	44000	0333-6853234	45.99	2024
3	Rohan Malik	Street 4	Lahore	Punjab	20544	0300-3468643	5534.87	2022
4	Saamia Zee	Main Street 2	Mianwali	Punjab	22276	0343-7531435	987.43	2023
5	Subhana Hashim	Street 6	Sargodah	Punjab	22334	0344-7532685	76.64	2020
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ALTER TABLE campus RENAME COLUMN CampusName TO C_name 0 row(s) returned 0.003 sec

- Change the column name LastName to L_name in Members table.

Limit to 1000 rows

```

1 • SELECT * FROM university.member;
2
3 • ALTER TABLE member
4   RENAME COLUMN LastName TO L_name;

```

MemberID	FirstName	L_name	Address	CampusPhone	CampusID	PositionID	ContractDuration
11	Hannah	Zoe	Attock	0303-8624579	3	100	5
22	Sumaira	Safeer	Bhakkar	0333-4696429	5	200	3
33	Rohan	Malik	Multan	0344-4685325	1	300	7
44	Subhana	Hashim	Faislabad	0300-6864356	4	400	2
55	Saamia	Zee	Lahore	0333-8735785	2	500	4
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

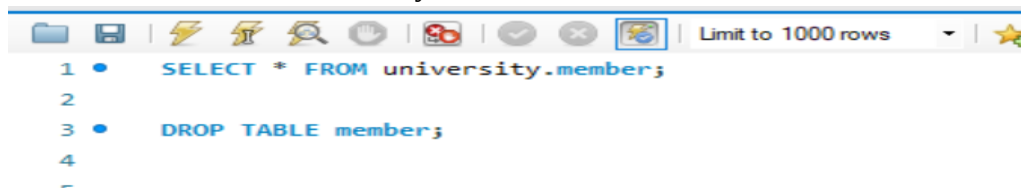
✓	120	20:48:02	SELECT * FROM university.member LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓	121	20:51:25	SELECT * FROM university.member LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Task 5:

RENAME: It is used to modify the name of the existing database object.

SYNTAX: RENAME TABLE old_relation_name TO new_relation_name;

- Delete the table Members from your database.

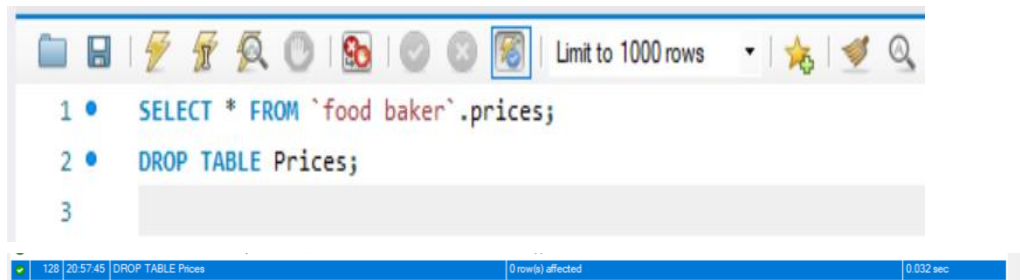


```

1 • SELECT * FROM university.member;
2
3 • DROP TABLE member;
4
-

```

- Delete the table Prices from your database.



```

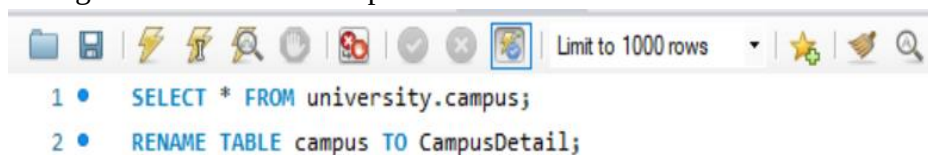
1 • SELECT * FROM `food baker`.prices;
2 • DROP TABLE Prices;
3

```

128 | 20:57:45 | DROP TABLE Prices | 0 row(s) affected | 0.032 sec

Task 6:

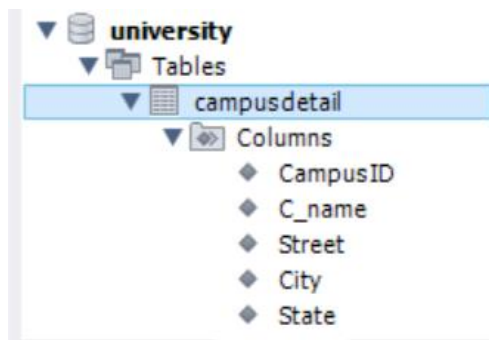
- Change the table name Campus to Cam



```

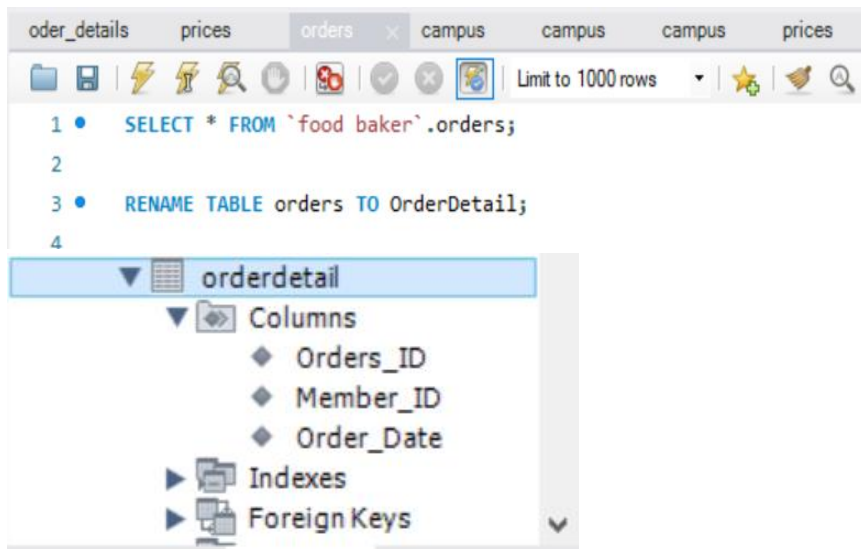
1 • SELECT * FROM university.campus;
2 • RENAME TABLE campus TO CampusDetail;

```



50 | 5/1/2023 | RENAME TABLE campus TO CampusDetail | 0 row(s) affected | 0.012 sec

- Change the table name Orders to OrderDeatil.



Post lab Tasks:

Create a table 'Book_register'.

Table Name: Schema: **university**

Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G	Default/Expression
ACC_NO	INT	☒	☒	☐	☐	☐	☐	☐	☐	
Book_Title	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
Pub_Name	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
Price	DECIMAL(10,2)	☐	☒	☐	☐	☐	☐	☐	☐	
book_registercol	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐	
		☐	☐	☐	☐	☐	☐	☐	☐	

Result Grid | Filter Rows: | Edit:

	ACC_NO	Book_Title	Pub_Name	Price	Author_Name
	34685	Linear Al...	Pearson	450...	David C. Lay
	45789	OS	Weliy	500...	Peter B. Galvin
	45795	Calculus	Cengage ...	650...	James Stewart
	56832	Mathema...	Oxford U...	350...	Timothy Gow...
	56890	DataBase	McGraw-Hill	230...	S. Sudarshan
*	NULL	NULL	NULL	NULL	NULL

ook_register1 x

Output

Action Output

#	Time	Action
1	22:25:28	SELECT * FROM university.book_register LIMIT 0, 1000

- Create a query for rename column ACC_No to Account_num.



- 1 • `SELECT * FROM university.book_register;`
- 2
- 3 • `ALTER TABLE book_register CHANGE ACC_NO Account_num INT;`

Result Grid | Filter Rows: | Edit: | Export/Import:

	Account_num	Book_Title	Pub_Name	Price	Author_Name
▶	34685	Linear Algebra	Pearson	4500.00	David C. Lay
	45789	OS	Weliy	5000.90	Peter B. Galvin
	45795	Calculus	Cengage Learning	6500.00	James Stewart
	56832	Mathematics	Oxford University Press	35000.00	Timothy Gowers
	56890	DataBase	McGraw-Hill	230.99	S. Sudarshan
*	NULL	NULL	NULL	NULL	NULL

book_register1 x

Output

2 22:55:33 SELECT * FROM university.book_register LIMIT 1000 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.000 sec \ 0.000 sec

- Drop the field PUB_NAME.



- 1 • `SELECT * FROM university.book_register;`
- 2 • `ALTER TABLE book_register DROP COLUMN Pub_Name;`

Result Grid | Filter Rows: | Edit: | Export/Import:

	Account_num	Book_Title	Price	Author_Name
▶	34685	Linear Algebra	4500.00	David C. Lay
	45789	OS	5000.90	Peter B. Galvin
	45795	Calculus	6500.00	James Stewart
	56832	Mathematics	35000.00	Timothy Gowers
	56890	DataBase	230.99	S. Sudarshan
*	NULL	NULL	NULL	NULL

37 22:55:35 ALTER TABLE book_register DROP COLUMN Pub_Name 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.046 sec

- Change the data type and size of the field PRICE Varchar (15).



- 1 • `SELECT * FROM university.book_register;`
- 2 • `ALTER TABLE book_register MODIFY Price VARCHAR(15);`

		Table Name: book_register	Schema: universi						
Column Name	Datatype	PK	NN	UQ	B	UN	ZF	AI	G
Account_num	INT	☒	☒	☐	☐	☐	☐	☐	☐
Book_Title	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐
Pub_Name	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐
Price	VARCHAR(15)	☐	☒	☐	☐	☐	☐	☐	☐
Author_Name	VARCHAR(45)	☐	☒	☐	☐	☐	☐	☐	☐
		☐	☐	☐	☐	☐	☐	☐	☐

✓	30	22:50:30	SELECT * FROM university.book_register LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✓	31	22:50:30	ALTER TABLE book_register MODIFY Price VARCHAR(15)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.094 sec